# Warren truss

Plane truss — analysis and CIRSOC 301 rod checks

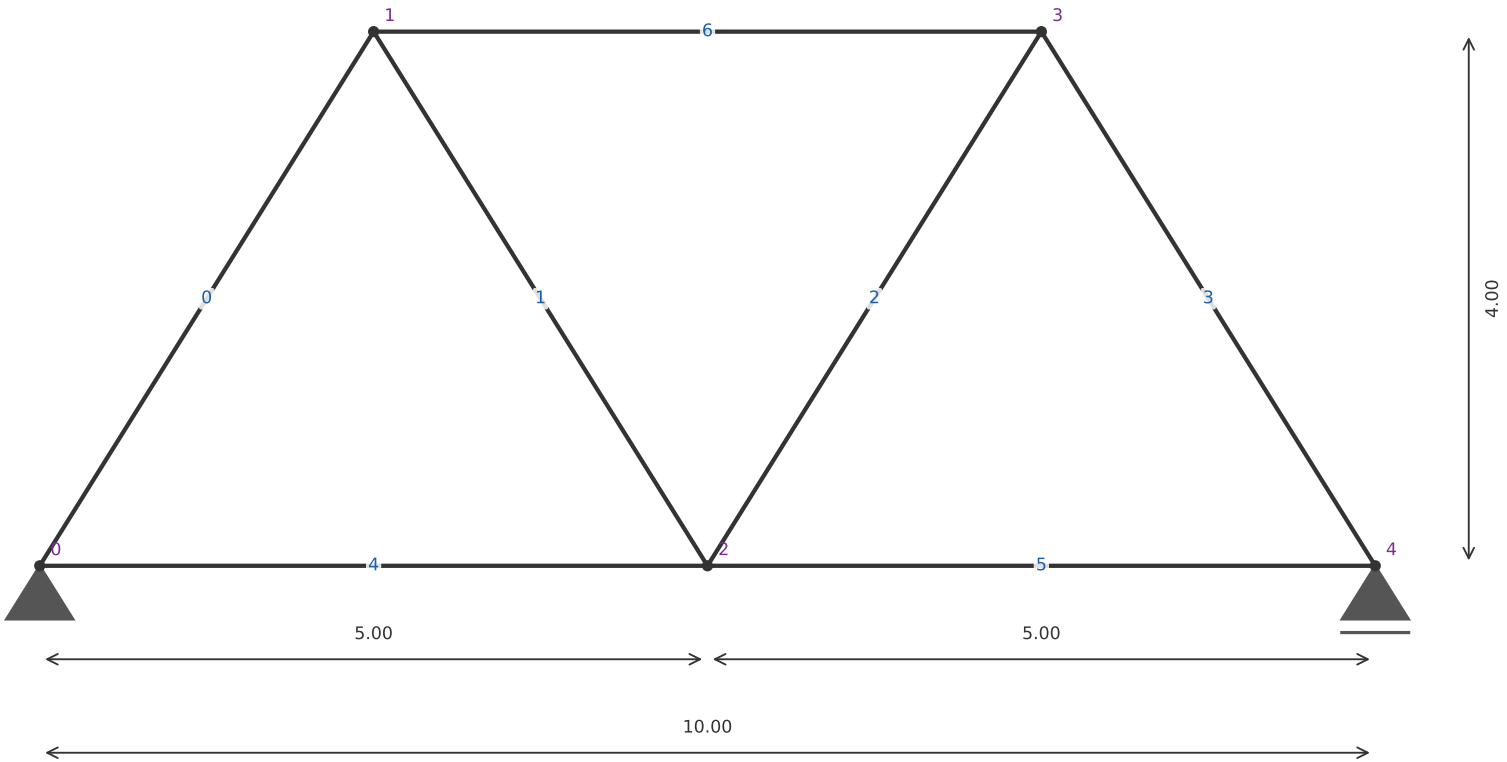
| | |
|---|---|
| Span | 10.00 m |
| Height | 4.00 m |
| Nodes / rods | 5 / 7 |
| Supports | pin at 0, rollerX at 4 |
| Loads | 3 nodal, total 120.0 kN |
| Family "Chord" | CHS 88.9x4.0 (F24 (CIRSOC) / A36) |
| Family "Web" | CHS 76.1x3.6 (F24 (CIRSOC) / A36) |
| Weight | Weighs 252 kg, carries 120.0 kN (49 × its own weight), FAILS: 2 rod(s) over capacity. |
| Deflection | 4.97 mm = L/2011, limit L/300: OK |
| Stability | Statically determinate (m + r − 2j = 7 + 3 − 2×5 = 0): statics alone gives every force. |
| Most used rod | rod 0 at 1.87 (elastic buckling (E3-3)) — OVER CAPACITY |
| Rods over capacity | 2 of 7 checked |
| Buckling | $\lambda_1$ = 0.68 — BELOW 1, it buckles before the full load is on |
| Shear and moment | none: every rod is pinned and loaded only at its nodes, so the axial force is the whole result |

Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

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7. Buckling
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9. Explain rod 0 and 3

Node numbers in purple, rod numbers in blue. Dimensions in m.




2 m · true to scale

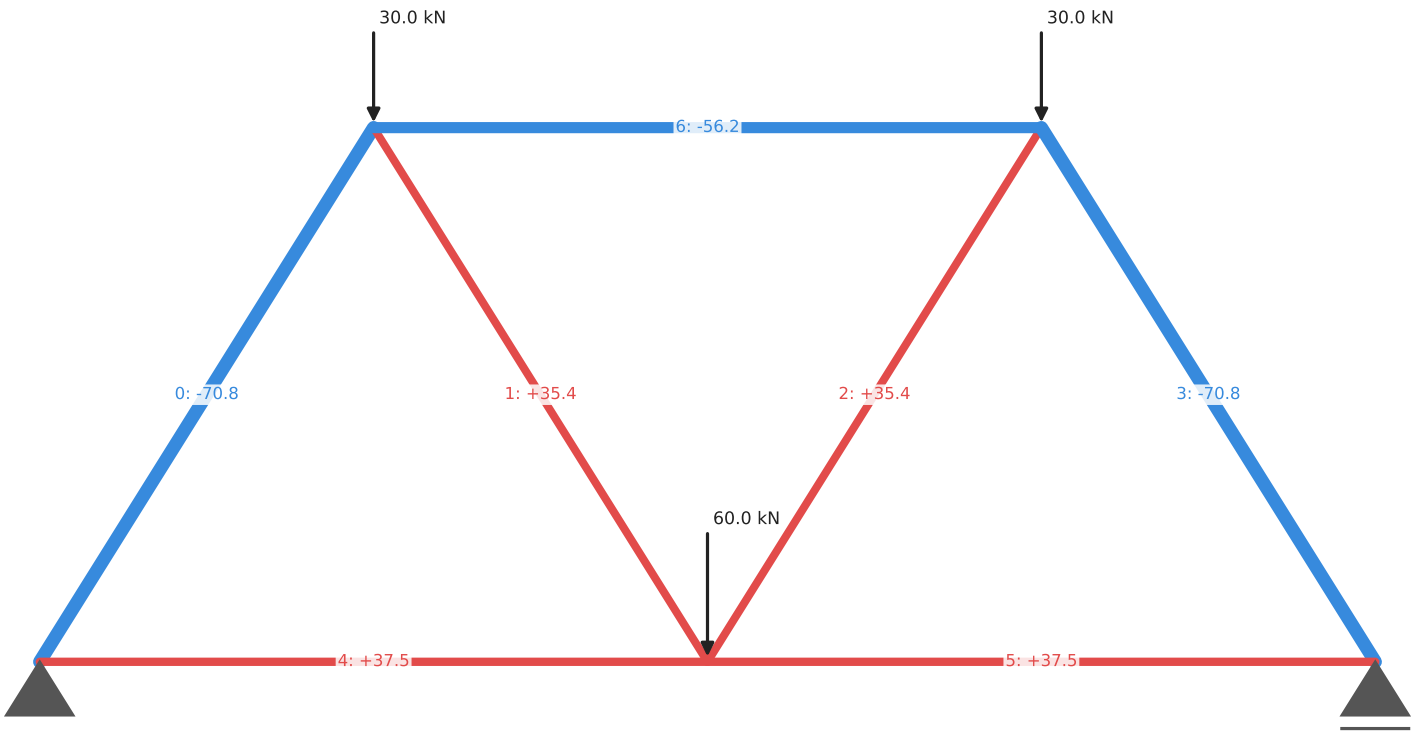
0 2

## Truss — Geometry



| MODEL | SIZE | UNITS | CODE | DATE |
|---|---|---|---|---|
| Warren truss | 5 nodes / 7 rods | m, kN, mm | CIRSOC 301 | 2026-10-12 02:24 |

Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 70.8).




2 m · true to scale

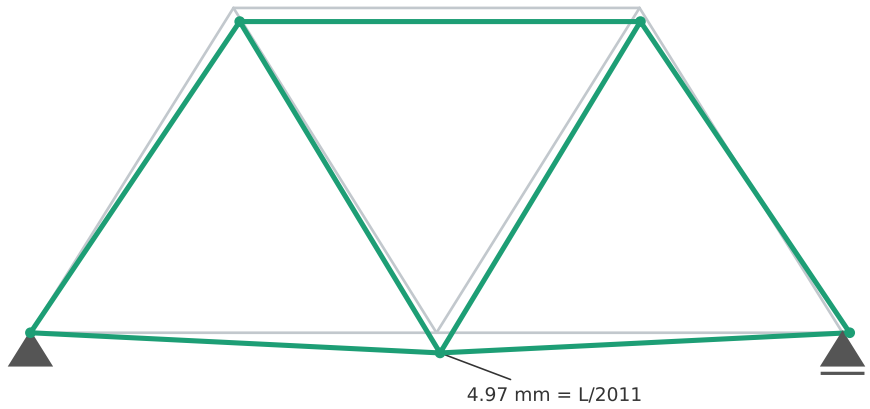
0 2

## Truss — Axial forces

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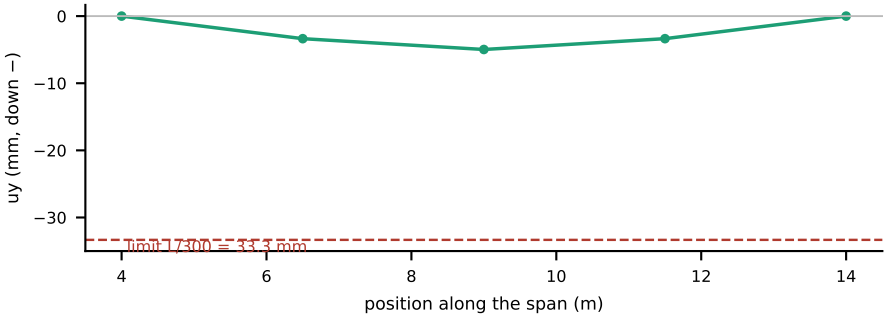
| MODEL | SIZE | UNITS | CODE | DATE |
|---|---|---|---|---|
| Warren truss | 5 nodes / 7 rods | m, kN, mm | CIRSOC 301 | 2026-10-12 02:24 |

Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50. Every rod is pinned, so every member stays straight: there is no curvature, and no contraflexure. Largest vertical deflection 4.97 mm against the limit L/300 = 33.3 mm: OK.




## Deflection along the span

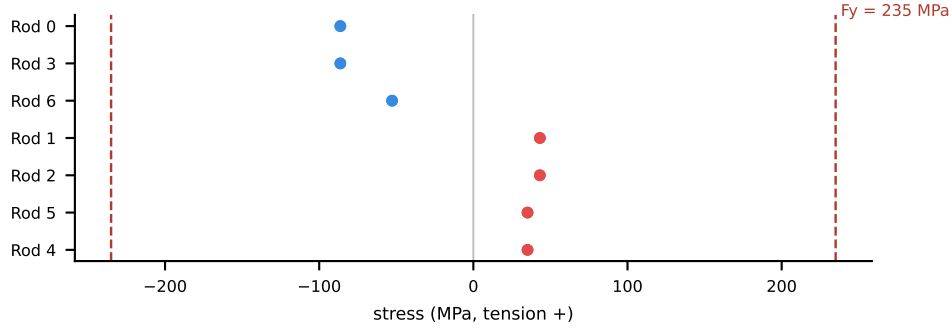
vertical movement of every joint




Largest 4.97 mm at joint 2 = L/2011. Within the limit.

## Fibre stress

N/A ± M·c/I at each rod's extreme fibres




Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

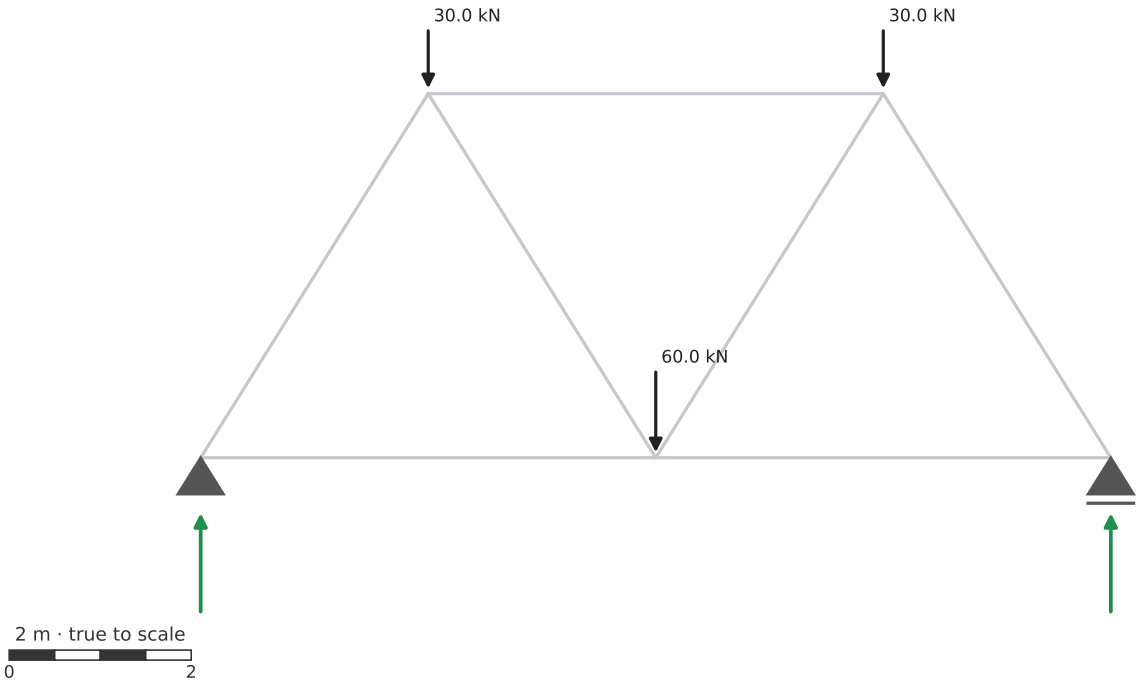
# Truss — Deflection and stress



| MODEL | SIZE | UNITS | CODE | DATE |
|---|---|---|---|---|
| Warren truss | 5 nodes / 7 rods | m, kN, mm | CIRSOC 301 | 2026-10-12 02:24 |

Support reactions (green arrows), as the supports push on the truss; Ry positive upwards. The sums balance the loads.

| Node | Type | Rx (kN) | Ry (kN) |
|---|---|---|---|
| 0 | pin | 0.00 | 60.00 |
| 4 | rollerX | 0.00 | 60.00 |
| Σ | | 0.00 | 120.00 |

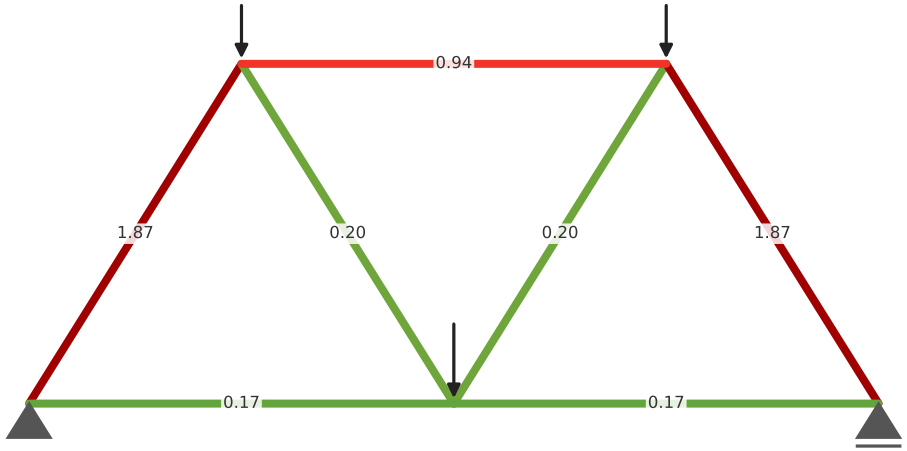



## Truss — Reactions

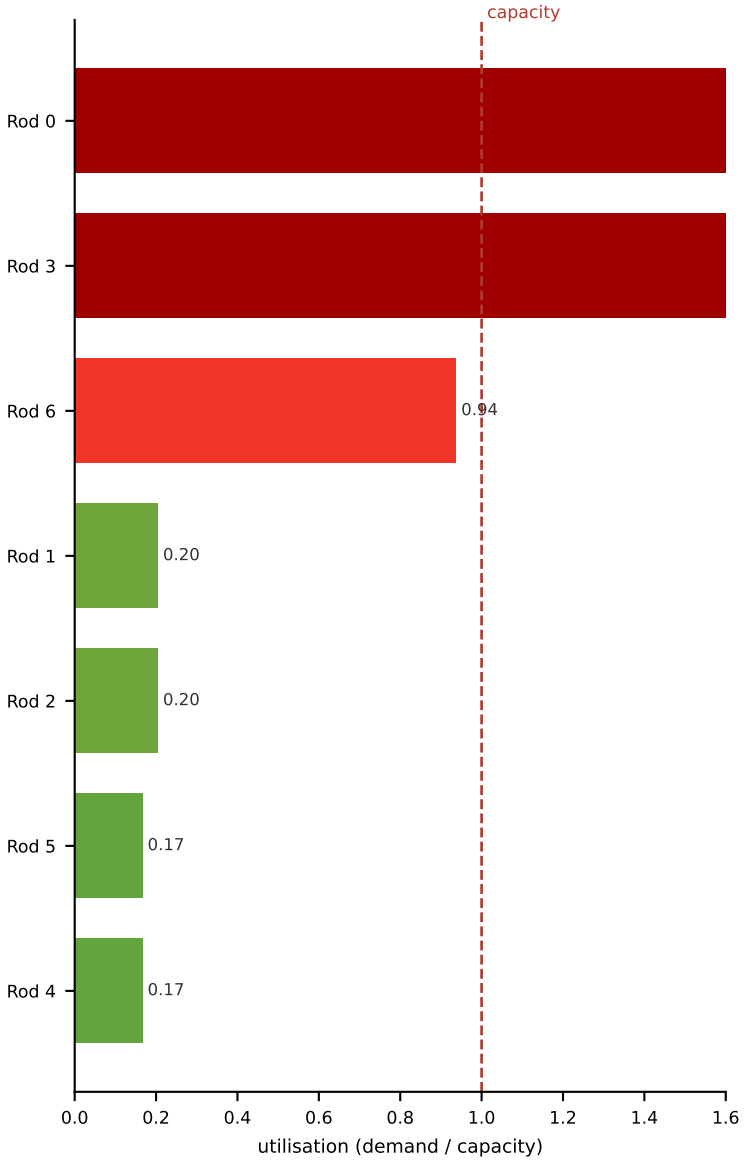


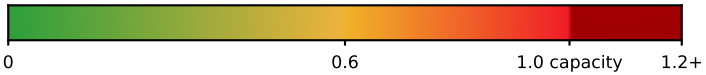
| MODEL | SIZE | UNITS | CODE | DATE |
|---|---|---|---|---|
| Warren truss | 5 nodes / 7 rods | m, kN, mm | CIRSOC 301 | 2026-10-12 02:24 |

Utilisation: demand over CIRSOC 301 capacity. Most used rod 0 at 1.87 (elastic buckling (E3-3)).  2 rod(s) OVER capacity: 0, 3.




2 m · true to scale
0 2




grey = no steel section, not checked

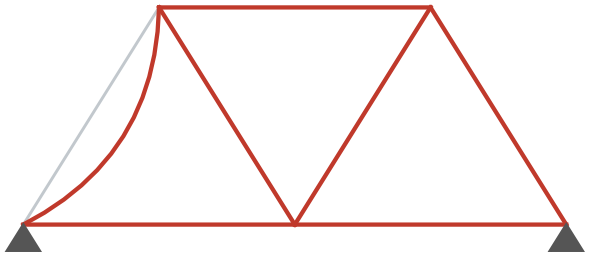
## Truss — Utilisation

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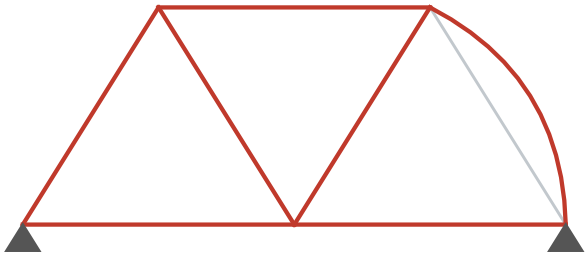
| MODEL | SIZE | UNITS | CODE | DATE |
|---|---|---|---|---|
| Warren truss | 5 nodes / 7 rods | m, kN, mm | CIRSOC 301 | 2026-10-12 02:24 |

λ is the factor on the load as it stands. $\lambda_1 = 0.68$ is BELOW 1: it buckles before the full load is on. 2 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

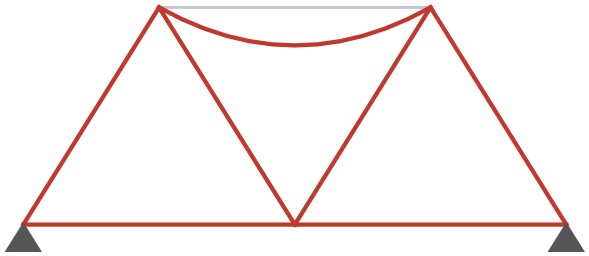
Mode 1: λ = 0.68
rod 0 bows

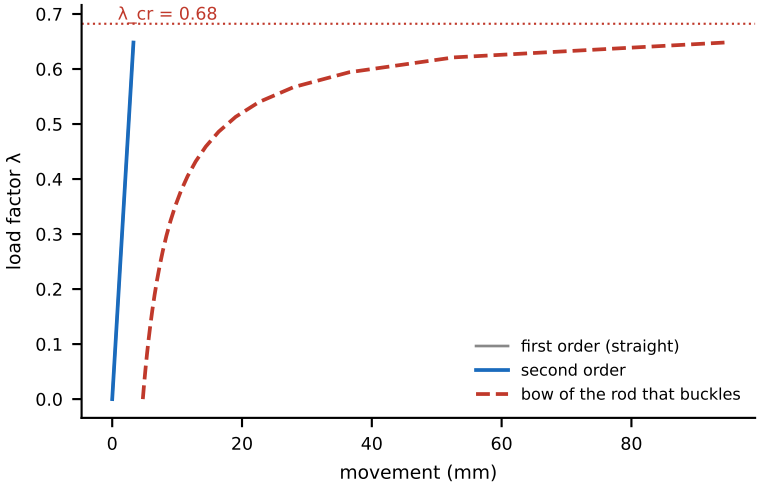


Mode 2: λ = 0.68
rod 3 bows



Mode 3: λ = 1.36
rod 6 bows






The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 4.72 mm in the first mode's shape.

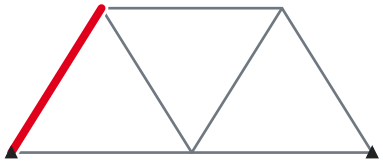
Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

**Truss — Buckling**



| MODEL | SIZE | UNITS | CODE | DATE |
|---|---|---|---|---|
| Warren truss | 5 nodes / 7 rods | m, kN, mm | CIRSOC 301 | 2026-10-12 02:24 |

| Rod | Nodes | Family | Section | L (m) | N (kN) | Governing check | Use | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0–1 | Web | CHS 76.1x3.6 | 4.72 | -70.75 | elastic buckling (E3-3) | 1.87 | OVER |
| 1 | 1–2 | Web | CHS 76.1x3.6 | 4.72 | +35.38 | normal stress (H.3.4) | 0.20 | OK |
| 2 | 2–3 | Web | CHS 76.1x3.6 | 4.72 | +35.38 | normal stress (H.3.4) | 0.20 | OK |
| 3 | 3–4 | Web | CHS 76.1x3.6 | 4.72 | -70.75 | elastic buckling (E3-3) | 1.87 | OVER |
| 4 | 0–2 | Chord | CHS 88.9x4.0 | 5.00 | +37.50 | normal stress (H.3.4) | 0.17 | OK |
| 5 | 2–4 | Chord | CHS 88.9x4.0 | 5.00 | +37.50 | normal stress (H.3.4) | 0.17 | OK |
| 6 | 1–3 | Chord | CHS 88.9x4.0 | 5.00 | -56.25 | elastic buckling (E3-3) | 0.94 | OK |



## Rod 0

utilisation 1.87  OVER

elastic buckling (E3-3)

```
Rod 0
=====
Steel, CIRSOC 301-2018 -- A = 820 mm², Fy = 235 MPa, L = 4.717 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,717 mm / 25.7 mm
      = 183.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 183.8²
      = 58.43 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 183.8 vs 4.71 √(200,000/235) = 137.4
      = elastic

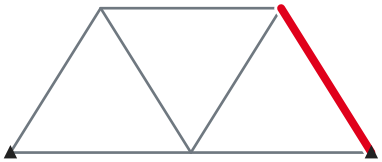
 4. Critical stress  [E3-3]
      Fcr = 0.877 · Fe
      = 0.877 × 58.43
      = 51.25 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 51.25 × 820
      = 37.82 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 70.75 / 37.82
      = 1.871

 7. Governs
      the largest ratio
      = elastic buckling (E3-3)
      = 1.871

utilisation 1.871 -- OVER CAPACITY
```



## Rod 3

utilisation 1.87  OVER

elastic buckling (E3-3)

```
Rod 3
=====
Steel, CIRSOC 301-2018 -- A = 820 mm², Fy = 235 MPa, L = 4.717 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,717 mm / 25.7 mm
      = 183.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 183.8²
      = 58.43 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 183.8 vs 4.71 √(200,000/235) = 137.4
      = elastic

 4. Critical stress  [E3-3]
      Fcr = 0.877 · Fe
      = 0.877 × 58.43
      = 51.25 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 51.25 × 820
      = 37.82 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 70.75 / 37.82
      = 1.871

 7. Governs
      the largest ratio
      = elastic buckling (E3-3)
      = 1.871

utilisation 1.871 -- OVER CAPACITY
```

## Truss — Explain rod 0 and 3



| MODEL | SIZE | UNITS | CODE | DATE |
|---|---|---|---|---|
| Warren truss | 5 nodes / 7 rods | m, kN, mm | CIRSOC 301 | 2026-10-12 02:24 |